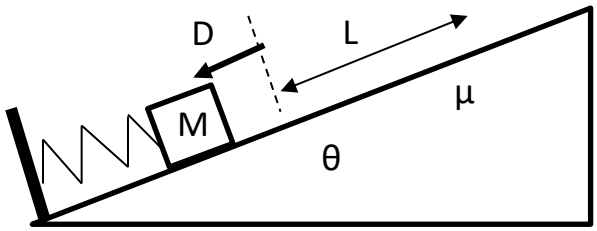
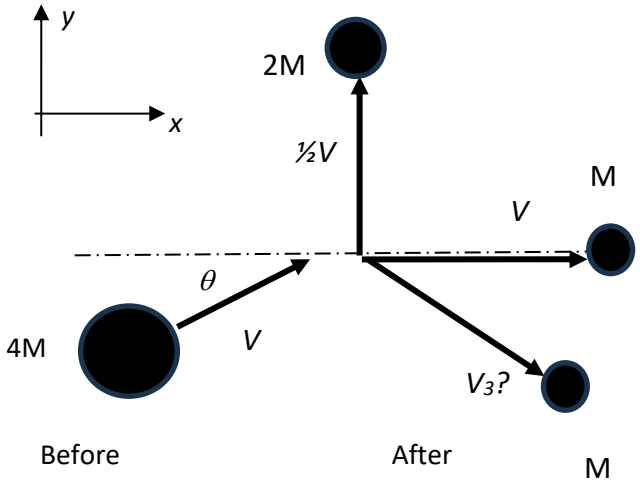
Physics 1135 **Test 2 Preparation Homework**

1. A box of mass M is on a rough incline that makes an angle θ with the horizontal. The coefficient of kinetic friction between the box and the incline is μ. The box is placed against a spring whose other end is secured to a wall at the lower end of the incline. The block is used to compress the spring a distance D and is then released from rest.



Derive an expression for the minimum spring constant k necessary to ensure that the box reaches a distance L up the incline **from the equilibrium position of the spring**. (Treat the box as a point mass.)

2. A firecracker of mass $4M$ is moving with speed V at an angle θ with the positive x-direction as shown in the figure and explodes. Immediately after the explosion, fragment 1 of mass $2M$ is moving with speed $\frac{1}{2}V$ in the positive y-direction. Fragment 2 of mass M continues with speed V in the positive x- direction. The third fragment moves at **unknown speed V_3** in an **unknown direction**.



In terms of system parameters, derive an expression for the velocity vector of the third fragment, in unit vector notation.

3. a) A projectile of mass m is shot directly away from the surface of a planet of mass M and radius R with an initial speed that equals **½ the escape speed** from the planet. Derive an expression for the maximum distance from the center of the planet the projectile reaches, in terms of R.

b) A satellite of mass m is put into a circular orbit around the planet of mass M and radius R. It orbits a distance $2R$ above the planet's surface. Derive an expression for total mechanical energy E of the satellite, in terms of G, m, M and R.

